Battery Design Calculation Sheet

Case ID: VBF-T6R1MIMO-CALC-001

Date: 2026-08-31

1. Basic Parameters

Parameter	Value	Unit	Source
Nominal Capacity	4.96	Ah	elec_e_1c_discharge.json
Nominal Voltage	3.97	V	elec_e_energy.json
Energy	17.6	Wh	elec_e_energy.json
Mass	43.5	g	elec_e_energy.json
Volume	20.6	mL	elec_e_energy.json
Volumetric ED	852.8	Wh/L	elec_e_energy.json
Gravimetric ED	404.7	Wh/kg	elec_e_energy.json
DC Resistance	0.148	mΩ	elec_e_energy.json
Power Density	634.6	kW/kg	elec_e_energy.json

2. Electrode Parameters

Layer	Thickness (µm)	Porosity	Density (g/cm^3)	Areal Density (g/m^2)
Positive Electrode	63	0.32	3.2	10.6
Negative Electrode	73.3	0.32	1.8	12.8
Separator	25	0.47	0.9	0.25
Positive CC	16	-	2.7	4.3
Negative CC	10	-	8.9	10.8

3. Performance Metrics

Metric	Target	Achieved	Pass/Fail	Source
Volumetric ED	>=950 Wh/L	852.8	FAIL	elec_e_energy.json
Voltage Plateau	>=4.1 V	3.97	FAIL	elec_e_energy.json
4C Plating	No plating	No plating	PASS	elec_e_4c_charge.json
Max Temperature	<=50°C	79	FAIL	elec_e_4c_charge.json
SEI Thickness	<=500 nm	482	PASS	elec_e_aging_100cyc.json